

Runyang (Ray) Shu

University of Texas at Austin
Department of Marketing, McCombs School of Business
Austin, TX 78705
✉ Runyang.Shu@mcombs.utexas.edu

Education

- August 2021–Present **PhD in Quantitative Marketing**, *University of Texas at Austin*, Texas, United States
- July 2019–June 2021 **Master of Science in Management**, *ESSEC Business School*, Paris, France
- October 2015–September 2019 **Bachelor's in Business Administration (Minor in English)**, *Beijing Foreign Studies University*, Beijing, China

Research Interests

- Methodological Machine Learning, Computer Vision
- Substantive Visual Marketing, Consumer Aesthetics, Picture Recommendation

Selected Works

- Seeing Is Believing: Improving Recommendations through Image Similarity between Consumers:** with Alain Lemaire - Under Review at *Management Science*
- Under Review This study introduces image similarity as a behavioral signal for recommender systems. Leveraging 5 M Yelp photos, we compute object-level and aesthetic similarity between user- and business-uploaded images (CLIP, VQ-VAE). Both signals predict ratings beyond structured attributes, text, and image quality; aesthetic similarity is especially powerful for ambience-centric venues. Incorporating similarity lifts out-of-sample AUC by 4.3% across models. Clustering images into interpretable themes isolates visual elements (e.g., “Sushi”, “Warm, Bright, Smooth”) that drive preference, and a controlled experiment confirms users favor restaurants whose images mirror their own. Findings support scalable, privacy-sensitive visual personalization.
- Works in Progress **Capturing Emotional Arc in Ad Videos:** with Rex Du and Alain Lemaire
- We develop an interpretable multimodal network that fuses second-by-second galvanic-skin-response, video, and audio through cross-attention to trace each ad’s emotional arc—the joint trajectory of arousal and valence. Leveraging clip-level valence labels and film-theory benchmarks, we project arcs onto six canonical narrative shapes (e.g., “rise–fall,” “steady rise”). These low-dimensional arc weights predict post-exposure liking and recall significantly better than models using content or physiology alone, while revealing which narrative patterns drive persuasion.

Works in Progress **How to Spur Adoption of a Platform-Specific Large Language Model? Evidence from a 6 500-Student Field Experiment on a Coding-Education Platform:** with Yixing Chen and Ananya Sen

We conduct a three-arm randomized email experiment on a 6,500-student coding-education platform to pinpoint which framing best triggers first-time use of the platform's in-house LLM assistant. Students receive (i) a minimal announcement (Control), (ii) a "Specs" message emphasizing model architecture and training data, or (iii) a "Human-Performance" message highlighting proven gains in coding speed and success. The primary outcome is adoption within 30 days; secondary metrics track total calls and eight-week retention.

Teaching Experience

2024 spring **Instructor**, *University of Texas at Austin*, Principles of Marketing, Undergraduate
Overall Instructor Rating: 4.9/5

Relevant Coursework

Marketing	Behavioral Decision Research (Raj Raghunathan), Marketing Research Methods (Rex Du), Marketing Models I (Raghunath S. Rao), Marketing Models II (Jason Duan), Marketing Management and Strategy (Leigh McAlister)
Economics	Microeconomic Theory I (Vasiliki Skreta), Microeconomic Theory II (Caroline Thomas), Econometrics I (Stephen Donald), Econometrics II (Brendan Kline), Structural Econometrics (Daniel Akerberg)
Machine Learning and Statistics	Online Learning and Optimization (Andrew Whinston), Optimization In Machine Learning (Nhat Ho)

Conference Attendance

2023 Marketing Science Conference and ISMS Doctoral Consortium, Miami, June
2024 Bass FORMS Conference, March

Working Experience

2019 **Research Assistant**, *UNESCO, Beijing*, Beijing, China
2020 **Research Assistant**, *ESSEC Business School*, Paris, France
2021 **Research Assistant**, *INSEAD Business School*, Paris, France

Honors

2021, 2022, 2023 Recruitment Graduate School College Fellowship
2021, 2022, 2023 McCombs School Fellowship

Certifications & Skills

Computer R, Python (Pytorch & Tensorflow), MATLAB, Stan

Language Proficient in English, Native in Mandarin, Conversational in French